

## 2-Edge connected dominating sets and 2-Connected dominating sets of a graph

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Published online: 21 August 2014  
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**Abstract** A  $k$ -connected (resp.  $k$ -edge connected) dominating set  $D$  of a connected graph  $G$  is a subset of  $V(G)$  such that  $G[D]$  is  $k$ -connected (resp.  $k$ -edge connected) and each  $v \in V(G) \setminus D$  has at least one neighbor in  $D$ . The  $k$ -connected domination number (resp.  $k$ -edge connected domination number) of a graph  $G$  is the minimum size of a  $k$ -connected (resp.  $k$ -edge connected) dominating set of  $G$ , and denoted by  $\gamma_k(G)$  (resp.  $\gamma'_k(G)$ ). In this paper, we investigate the relation of independence number and 2-connected (resp. 2-edge-connected) domination number, and prove that for a graph  $G$ , if it is 2-edge connected, then  $\gamma'_2(G) \leq 4\alpha(G) - 1$ , and it is 2-connected, then  $\gamma_2(G) \leq 6\alpha(G) - 3$ , where  $\alpha(G)$  is the independent number of  $G$ .

**Keywords** Connected dominating set · Dominating set · Independent set

### 1 Introduction

A wireless sensor networks is composed of many sensors to monitor physical or environmental conditions, and to cooperatively pass their data through the networks. Thus such a network has been widely used in many industries such as environment monitoring, disaster recovery, health applications, etc. To keep nodes in a wireless network being able to communicate each other, the sensor network is required to have certain connectivity. Such a task is called topological control. The virtual backbone

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has been introduced to involve in topological control for wireless sensor networks to reduce the utilization of resource. Usually, A network is modeled as a graph, and a virtual backbone is formulated as a connected dominating set. We refer to [Pardalos \(2013\)](#) for applications of connected dominating set.

Sometimes, two nodes cannot communicate due to electromagnetic interference, what's worse, some nodes may fail because of various reasons. So, a dominating set with higher connectivity is expected. The problem concerning domination of graphs (or networks) is a major area in graph theory that attracted a large number of researchers and generate a wealth of important achievements in the past few decades.

All graphs considered are finite, undirected, simple. We follow the notation and terminology of Bondy and Murty ([2008](#)), unless otherwise stated. A graph is *k-connected* if deleting any set of at most  $k - 1$  vertices results in a connected graph with at least two vertices, or a trivial graph. A graph is *k-edge connected* if deleting any set of at most  $k - 1$  edges results in a connected graph. Let  $G = (V, E)$  be a graph and  $D \subseteq V$ . The subgraph of  $G$  induced by  $D$ , denoted by  $G[D]$ , is the graph with  $D$  as the vertex set, in which two vertices are adjacent if and only if they are adjacent in  $G$ .  $D$  is an *independent set* if no two vertices of  $D$  are adjacent in  $G$ . The independence number of  $G$ , denoted by  $\alpha(G)$ , is the maximum cardinality of an independent set of  $G$ . We call  $D$  a *dominating set* of  $G$  if every vertex in  $V \setminus D$  has a neighbor in  $D$ . Furthermore, if  $G[D]$  is *k-connected* (resp. *k-edge connected*),  $D$  is called a *k-connected* (resp. *k-edge connected*) dominating set. The *k-connected domination number* (resp. *k-edge connected domination number*) of a graph  $G$ , denoted by  $\gamma_k(G)$  (resp. by  $\gamma'_k(G)$ ) is the minimum cardinality of a *k-connected* (resp. *k-edge connected*) dominating set. Clearly, a graph  $G$  has a *k-connected* (resp. *k-edge connected*) dominating set if  $G$  is *k-connected* (resp. *k-edge connected*). But a graph having a *k-connected* (resp. *k-edge connected*) dominating set need not to be *k-connected* (resp. *k-edge connected*).

It is clear that  $\gamma'_0(G) = \gamma_0(G)$  and  $\gamma'_1(G) = \gamma_1(G)$  for any graph  $G$ . Note that  $\gamma_0(G)$  is the domination number of  $G$  and  $\gamma_1(G)$  is the connected domination number of  $G$ . There are some upper bounds for the connected domination number of a connected graph on  $n$  vertices in terms of its minimum degree in literature. In [Caro and Yuster \(2003\)](#), it was conjectured by Caro et al. that, if  $G$  is a *k-connected* graph with  $n$  vertices and minimum degree  $\delta$ , then  $\gamma_k(G) \leq n \frac{\ln \delta}{\delta} (1 + o_\delta(1))$ , where  $k$  is a positive integer. The validity of this conjecture is verified for  $k = 1$  in [Caro et al. \(2000\)](#) and for  $k = 2$  in [Caro and Yuster \(2003\)](#).

The theory of connected domination of graphs has important applications in communication and computer networks, especially its role as a virtual backbone in wireless networks, see Du and Wan ([2013](#)). [Haynes et al. \(1997a, b\)](#) published two monographs concerning domination in graphs, and recently [Chellali et al. \(2012\)](#) presented a survey paper concerning dominating sets and independent sets. We refer to [Arseneau et al. \(1997\)](#), [Bo and Liu \(1996\)](#), [Caro and Yuster \(2003\)](#), [Colbourn and Stewart \(1990\)](#), [Li and Zhang \(2010\)](#), [Moscarini \(1993\)](#), [Sampathkumar and Walikar \(1979\)](#), [White et al. \(1985\)](#) for more results concerning connected dominating sets.

An interesting application of the connected domination of graphs is in minor theory. The well-known Hadwiger's conjecture states that if  $\chi(G) \geq k$ , then  $G$  contains a  $K_k$ -minor, where  $\chi(G)$  denotes the chromatic number of  $G$ . Since

$$\alpha(G)\chi(G) \geq n$$

for a graph  $G$  on  $n$  vertices, Hadwiger's conjecture implies that any graph  $G$  on  $n$  vertices has a  $K_{\lceil \frac{n}{\alpha(G)} \rceil}$ -minor. [Duchet and Meyniel \(1982\)](#) established the following relation between the connected domination number and the independence number of a connected graph, and by applying this result, they proved that any graph  $G$  on  $n$  vertices has a  $K_{\lceil \frac{n}{2\alpha(G)-1} \rceil}$ -minor.

**Theorem 1** ([Duchet and Meyniel 1982](#)) *For any connected graph  $G$ ,  $\gamma_1(G) \leq 2\alpha(G) - 1$ .*

In some sense, the above theorem of Duchet and Meyniel is related to the following conjecture in the combinatorial optimization.

**Conjecture 1** ([Wu et al. 2010](#)) *For any connected unit disk graph  $G$ ,  $\alpha(G) \leq 3\gamma_1(G) + 2$ .*

There are a number of papers devoted to the relation of the independence number and the connected domination number of unit disk graphs, for instance, [Li et al. \(2011\)](#), [Wan et al. \(2008\)](#), [Wu et al. \(2006\)](#). Best known result on Conjecture 1 is  $\alpha(G) \leq 3.4306\gamma_1(G) + 4.8185$  obtained by [Li et al. \(2011\)](#). So, combining this with Theorem 1, for a connected unit disk graph  $G$ ,

$$0.5\gamma_1(G) + 0.5 \leq \alpha(G) \leq 3.4306\gamma_1(G) + 4.8185$$

We refer to [Wu et al. \(2010\)](#) for more relevant works concerning domination and packing on wireless networks.

As far as we know, there are no known results on the relation between the independence number and the 2-connected (resp. the 2-edge connected) domination number of a graph. We extend the result of [Duchet and Meyniel \(1982\)](#) as follows.

**Theorem 2** *If  $G$  is a 2-edge connected graph with an induced cycle of length  $c$ , then*

$$\gamma'_2(G) \leq \begin{cases} 4\alpha(G) - c, & \text{if } c \text{ is even} \\ 4\alpha(G) - c + 2, & \text{if } c \text{ is odd,} \end{cases}$$

*in particular,  $\gamma'_2(G) \leq 4\alpha(G) - 1$ .*

**Theorem 3** *If  $G$  is a 2-connected graph with an induced cycle of length  $c$ ,*

$$\gamma_2(G) \leq \begin{cases} 6\alpha(G) - 2c, & \text{if } c \text{ is even,} \\ 6\alpha(G) - 2c + 3, & \text{if } c \text{ is odd,} \end{cases}$$

in particular,  $\gamma_2(G) \leq 6\alpha(G) - 3$ .

**Theorem 4** If  $G$  is a 2-connected triangle-free graph with an induced cycle of length  $c$ ,

$$\gamma_2(G) \leq \begin{cases} 5\alpha(G) - \frac{3c}{2}, & \text{if } c \text{ is even,} \\ 5\alpha(G) - \frac{3c-1}{2} + 2, & \text{if } c \text{ is odd,} \end{cases}$$

in particular,  $\gamma_2(G) \leq 5\alpha(G) - 5$ .

The bounds of Theorems 2, 3, and 4 can be attained by the cycle  $C_n$  on  $n \geq 4$  vertices. The bounds of Theorems 2 and 3 can also be attained by  $K_n$  on  $n$  vertices.

## 2 2-Edge connected dominating set

For two vertices  $u, v \in V(G)$ , the distance  $d_G(u, v)$  between  $u$  and  $v$  is the number of edges in a shortest path connecting  $u$  and  $v$  in  $G$ . In general, for  $X \subseteq V(G)$  and  $Y \subseteq V(G)$ , the distance  $d_G(X, Y)$  between  $X$  and  $Y$  is  $\min\{d_G(x, y) : x \in X, y \in Y\}$ . Thus  $d_G(X, Y) = d_G(Y, X)$ . If  $Y = \{y\}$  for some vertex  $y \in V(G)$ , we simply write  $d_G(X, y)$  instead of  $d_G(X, \{y\})$ .

For any integer  $k \geq 1$ , the  $k$ -step open neighborhood  $N_G^k(X)$  is  $\{y \in V(G) : d(X, y) = k\}$ . We simply write  $N_G(X)$  for  $N_G^1(X)$  and  $N_G^k(x)$  for  $N_G^k(\{x\})$ .

Note that all trees do not have a 2-edge connected dominating set. A *pendent vertex* of a graph  $G$  is a vertex of degree one. A criterion for a general graph to admit a 2-edge connected dominating set is given bellow.

**Proposition 1** Let  $G$  be a connected graph and  $G'$  the graph obtained from  $G$  by deleting its all pendent vertices. Then  $G$  has a 2-edge connected dominating set if and only if  $G'$  is 2-edge connected.

*Proof* If  $G'$  is 2-edge connected, then  $V(G')$  is a 2-edge connected dominating set of  $G$ . Next we show its necessity by contradiction. Suppose that  $G'$  has a cut edge  $e = x_1x_2$ , and let  $G'_1$  and  $G'_2$  be two component of  $G' - e$  with  $x_i \in V(G'_i)$  for  $i = 1, 2$ . If  $D$  is a 2-edge connected dominating set of  $G$ , then  $D \subseteq V(G'_1)$  or  $D \subseteq V(G'_2)$ . Without loss of generality, let  $D \subseteq V(G'_1)$ . Since  $D$  is a dominating set of  $G$ ,  $G'_2$  must be the trivial graph. Thus  $V(G'_2) = \{x_2\}$  and  $x_2$  is a pendent vertex of  $G$ . But, this contradicts our assumption that  $G'$  is obtained from  $G$  deleting its all pendent vertices.  $\square$

The concept of the abstemious path plays a key role in proving our main results. For a graph  $G$ ,  $D \subseteq V(G)$  and  $x \in V(G) \setminus D$ , a path  $P$  joining  $x$  and  $D$  is *abstemious* if  $d(y, D) \leq d(x, D)$  for any  $y \in V(P)$ . We say that  $x$  is an *abstemious* vertex for  $D$  if there exist two edge-disjoint abstemious paths joining  $x$  and  $D$ . For an abstemious

vertex  $x$ , we use  $\ell_2(x, D)$  to denote the minimum integer such that there exists a pair of edge-disjoint abstemious paths  $P$  and  $Q$  satisfying  $\ell(P) + \ell(Q) = \ell_2(x, D)$ , where  $\ell(P)$  and  $\ell(Q)$  are the length of  $P$  and  $Q$ , respectively.

The following example  $G$  shows that  $\ell_2(G)$  can be arbitrarily large. Let  $C$  be a cycle of length at least three. By identifying the two end vertices of  $P = v_0 v_1 \dots v_k$  with two different vertices of  $C$ , we obtain a graph  $G$ . Let  $D = V(C)$ . If  $k$  is even, then  $v_{\frac{k}{2}}$  is the only abstemious vertex with respect to  $D$  and  $G$ , and  $\ell_2(v_{\frac{k}{2}}, D) = k$ ; if  $k$  is odd, then  $v_{\lfloor \frac{k}{2} \rfloor}$  and  $v_{\lceil \frac{k}{2} \rceil}$  are two abstemious vertices with respect to  $D$  and  $G$ , and  $\ell_2(v_{\lfloor \frac{k}{2} \rfloor}, D) = \ell_2(v_{\lceil \frac{k}{2} \rceil}, D) = k$ .

**Proof of Theorem 2** Let  $C$  be an induced cycle of length  $c$  in  $G$ . Set  $I_0$ : the set of nonadjacent vertices of cardinality  $\lfloor c/2 \rfloor$  in  $C$  and  $D := V(C)$ . If  $D$  is a dominating set of  $G$ , we are done, since  $G[D]$  is 2-edge connected. Otherwise, there exists a vertex  $x$  with  $d(x, D) \geq 2$ . Let  $X = \{x : d(x, D) \geq 2, \text{ and } x \text{ is an abstemious vertex with respect to } D \text{ and } G\}$ .  $\square$

**Claim 1**  $X \neq \emptyset$ .

**Proof** Let  $u$  be a vertex with  $d_G(u, D) \geq 2$ . Since  $G$  is 2-edge connected, there exist two edge-disjoint paths  $P$  and  $Q$  joining  $u$  and  $D$ . Pick a vertex  $x \in V(P) \cup V(Q)$  such that  $d_G(x, D) = \max\{d_G(v, D) : v \in V(P) \cup V(Q)\}$ . It is clear that there exist two edge-disjoint paths  $P'$  and  $Q'$  joining  $x$  and  $D$  such that  $E(P') \cup E(Q') = E(P) \cup E(Q)$ . By the choice of  $x$ , for any vertex  $y \in V(P') \cup V(Q')$ ,  $d(y, D) \leq d(x, D)$ . Hence  $P'$  and  $Q'$  are a pair of abstemious paths joining  $x$  and  $D$  and  $x$  is an abstemious vertex for  $D$ . So  $x \in X$ , implying that  $X \neq \emptyset$ .  $\square$

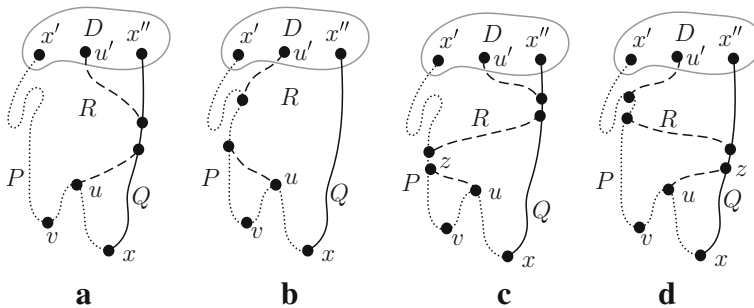
For the sake of clarity, we need an additional notations. For  $D \subseteq V(G)$ ,  $x \in V(G) \setminus D$ , let  $P$  be path joining  $x$  and  $D$ . For  $u, v \in V(P)$ ,  $uPv$  is the subpath of  $P$  starting from  $u$  and ending at  $v$ .

**Claim 2** Choose  $x \in X$  such that  $\ell_2(x, D) = \min\{\ell_2(v, D) : v \in X\}$  and a pair of abstemious paths  $P$  and  $Q$  joining  $x$  and  $D$  such that  $\ell(P) + \ell(Q) = \ell_2(x, D)$ . If  $k = d_G(x, D)$ , then

- (i) For any  $y, z \in V(P)$ , if  $\ell(x'Py) \leq \ell(x'Pz)$ , then  $d_G(y, D) \leq d_G(z, D)$ , where  $x'$  is the other end of  $P$  other than  $x$ ;
- (ii)  $(V(P) \setminus \{x\}) \cap N_G^k(D) = \emptyset$  or  $(V(Q) \setminus \{x\}) \cap N_G^k(D) = \emptyset$ ;
- (iii) if  $(V(P) \setminus \{x\}) \cap N_G^k(D) \neq \emptyset$ , then  $|(V(P) \setminus \{x\}) \cap N_G^k(D)| = 1$ , and the unique element of  $(V(P) \setminus \{x\}) \cap N_G^k(D)$  is adjacent to  $x$  on  $P$ ;
- (iv)  $|V(P) \cap N_G^i(D)| = |V(Q) \cap N_G^i(D)| = 1$  for any  $i \in \{2, \dots, k-1\}$ ;
- (v) if  $(V(P) \cup V(Q)) \setminus \{x\} \neq \emptyset$ , then  $(V(P) \cup V(Q)) \setminus \{x\} \subseteq N(D)$  and  $|(V(P) \cup V(Q)) \setminus \{x\}| = 1$ ;
- (vi)  $|(V(P) \cup V(Q)) \cap N(D)| \leq 2$ .

**Proof** Let us denote by  $x''$  the other end of  $Q$  other than  $x$ .

We show (i) by contradiction. Suppose that  $u, v$  are a pair of vertices of  $P$  such that (a)  $\ell(x'Pv) \leq \ell(x'Pu)$  and  $d_G(v, D) > d_G(u, D)$ , (b)  $d_G(v, D) \geq d_G(w, D) \geq$



**Fig. 1** Illustration for the proof of (i) of Claim 2

$d_G(u, D)$  for any  $w \in V(yPz)$ , (c)  $\ell(vPu)$  is as large as possible. In addition, subject to (a), (b) and (c),  $u, v$  has an additional property that  $d_G(v, D)$  is as large as possible.

Since  $P$  and  $Q$  is a pair of abstemious paths joining  $x$  and  $D$ , it follows that  $d_G(v, D) \leq d_G(x, D)$ . Let  $R$  be a path of length  $d_G(u, D)$  joining  $u$  and  $D$ , and let  $u'$  be the end of  $R$  in  $D$ . We consider the follow two cases.

*Case 1.*  $E(R) \cap E(P) = \emptyset$  or  $E(R) \cap E(Q) = \emptyset$ .

If  $E(R) \cap E(P) = \emptyset$  as shown in Fig. 1a, then the paths  $R' = vPuRu'$  and  $P' = vPx'$  are a pair of abstemious edge-disjoint paths joining  $v$  and  $D$  such that  $\ell(R') + \ell(P') = \ell(uPx') + \ell(R) < \ell(P) + \ell(Q)$ , a contradiction. If  $E(R) \cap E(Q) = \emptyset$  as shown in Fig. 1b, then the paths  $Q$  and  $P' = xPuRu'$  are a pair of abstemious edge-disjoint paths joining  $x$  and  $D$  such that  $\ell(Q) + \ell(P') = \ell(Q) + \ell(xPu) + \ell(R) < \ell(Q) + \ell(xPu) + \ell(uPx') = \ell(P) + \ell(Q)$ , a contradiction.

*Case 2.*  $E(R) \cap E(P) \neq \emptyset$  and  $E(R) \cap E(Q) \neq \emptyset$

*Subcase 2.1.*  $R$  first intersect  $P$  than  $Q$  as shown in Fig. 1c.

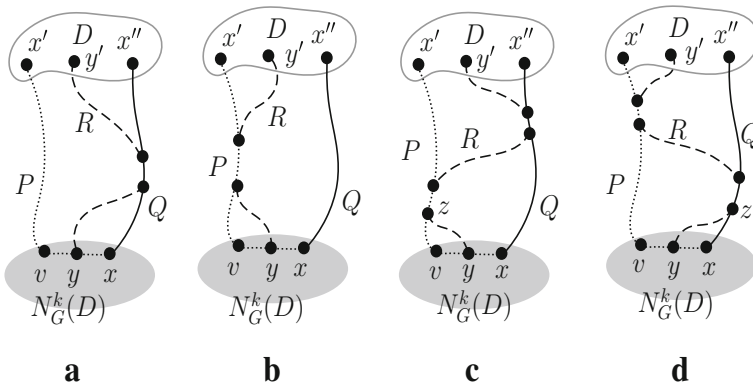
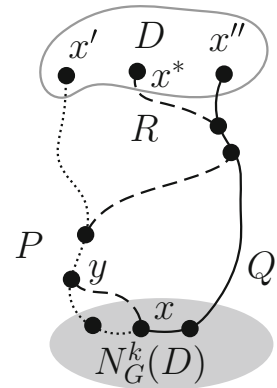
Let  $z$  be the common vertex of  $R$  and  $P$  when they first intersect. Then  $P' = xPuRzPx'$  and  $Q$  are a pair of abstemious edge-disjoint paths joining  $x$  and  $D$  such that  $\ell(P') + \ell(Q) = \ell(xPu) + \ell(uRz) + \ell(zPx') + \ell(Q) < \ell(xPu) + \ell(uPz) + \ell(zPx') + \ell(Q) = \ell(P) + \ell(Q)$ , a contradiction.

*Subcase 2.2.*  $R$  first intersect  $Q$  than  $P$  as shown in Fig. 1d.

Let  $z$  be the common vertex of  $R$  and  $P$  when they first intersect. Then  $P' = vPx'$  and  $Q' = vPuRzQx''$  are a pair of abstemious edge-disjoint paths joining  $v$  and  $D$  such that  $\ell(P') + \ell(Q') = \ell(vPx') + \ell(vPu) + \ell(uRz) + \ell(zQx'') < \ell(vPx') + \ell(vPu) + (\ell(uPx) + \ell(xQz)) + \ell(zQx'') = \ell(P) + \ell(Q)$ , a contradiction.

(ii) By contradiction, suppose that  $(V(P) \setminus \{x\}) \cap N_G^k(D) \neq \emptyset$  and  $(V(Q) \setminus \{x\}) \cap N_G^k(D) \neq \emptyset$ . Take a path  $R$  of length  $k$  joining  $x$  and  $D$ . Let  $x^*$  be the end of  $R$  in  $D$ . Clearly,  $R \neq P$  and  $R \neq Q$ . If  $E(R) \cap E(P) = \emptyset$  or  $E(R) \cap E(Q) = \emptyset$ , then  $R$  and one of  $P$  and  $Q$ , say  $P$ , are a pair of abstemious edge-disjoint paths joining  $x$  and  $D$  such that  $\ell(P) + \ell(R) < \ell(P) + \ell(Q)$ , a contradiction. So assume that  $E(R) \cap E(P) \neq \emptyset$  and  $E(R) \cap E(Q) \neq \emptyset$ . Without loss of generality, assume that  $R$  first intersect  $P$  than  $Q$  when it moves along  $R$  from  $x$  to  $D$ , and let  $y$  be the common vertex of  $R$  and  $P$  when they first intersect, see Fig. 2 for details. It is clear that  $P' = xRyPx'$  and  $Q$  are a pair of abstemious edge-disjoint paths joining  $x$  and  $D$

**Fig. 2** Illustration for the proof of (ii) of Claim 2



**Fig. 3** Illustration for the proof of (iii) of Claim 2

with  $\ell(P') + \ell(Q) = \ell(xRy) + \ell(yPx') + \ell(Q) \leq \ell(xPy) - 1 + \ell(yPx') + \ell(Q) = \ell(P) + \ell(Q) - 1 < \ell(P) + \ell(Q)$ . This proves (ii).

(iii) By contradiction, suppose that  $|V(P) \setminus \{x\} \cap N_G^k(D)| \geq 2$ . Let  $v$  be the vertex in  $V(P) \cap N_G^k(D)$  such that  $\ell(vPx)$  is maximized. For any  $w \in V(vPx)$ , we have that  $w \in N_G^k(D)$  by (i). Let  $y$  be the neighbor of  $x$  on the path  $xPv$ . Let  $R$  be a path of length  $d_G(y, D)$  joining  $y$  and  $D$ , and let  $y'$  be the end of  $R$  in  $D$ .

*Case 1.*  $E(R) \cap E(P) = \emptyset$  or  $E(R) \cap E(Q) = \emptyset$ .

If  $E(R) \cap E(P) = \emptyset$  as shown in Fig. 3a, then the paths  $R' = vPyRy'$  and  $P' = vPx'$  are a pair of abstemious edge-disjoint paths joining  $v$  and  $D$  such that  $\ell(P') + \ell(R') = \ell(vPx') + \ell(vpy) + \ell(R) \leq \ell(yPx') + \ell(R) < \ell(P) + \ell(Q)$ , a contradiction. If  $E(R) \cap E(Q) = \emptyset$  as shown in Fig. 3b, then the paths  $Q$  and  $P' = xPyRy'$  are a pair of abstemious edge-disjoint paths joining  $x$  and  $D$  such that  $\ell(P') + \ell(Q) = \ell(xPy) + \ell(yRy') + \ell(Q) < \ell(xPy) + \ell(yPx') + \ell(Q) = \ell(P) + \ell(Q)$ , a contradiction.

*Case 2.*  $E(R) \cap E(P) \neq \emptyset$  and  $E(R) \cap E(Q) \neq \emptyset$

*Subcase 2.1*  $R$  first intersect  $P$  than  $Q$  as shown in Fig. 3c.

Let  $z$  be the common vertex of  $R$  and  $P$  when they first intersect. Then  $P' = xPyRzPx'$  and  $Q$  are a pair of abstemious edge-disjoint paths joining  $x$  and  $D$  such that  $\ell(P') + \ell(Q) = \ell(xPy) + \ell(yRz) + \ell(zPx') + \ell(Q) < \ell(xPy) + \ell(yPz) + \ell(zPx') = \ell(P) + \ell(Q)$ , a contradiction.

*Subcase 2.2.  $R$  first intersect  $Q$  than  $P$  as shown in Fig. 3d.*

Let  $z$  be the common vertex of  $R$  and  $Q$  when they first intersect. Then  $P' = vPx'$  and  $Q' = vPyRzQx''$  are a pair of abstemious edge-disjoint paths joining  $v$  and  $D$  such that  $\ell(P') + \ell(Q') = \ell(vPx') + \ell(vPy) + \ell(yRz) + \ell(zQx'') < \ell(vPx') + \ell(vPy) + (\ell(yPx) + \ell(xQz)) + \ell(zQx'') = \ell(P) + \ell(Q)$ , a contradiction.

Next we show (iv). Suppose (iv) is false and let  $m$  be the maximum integer among  $\{2, \dots, k-1\}$  such that  $|V(P) \cap N_G^m(D)| \geq 2$ . Let  $y$  and  $z$  be the two vertices in  $V(P) \cap N_G^m(D)$  such that  $\ell(xPy)$  is maximized and  $\ell(xPz)$  is minimized. Let  $R$  be a path of length  $d_G(y, D)$  joining  $y$  and  $D$ . Let us denote by  $x'$  and  $x''$  the ends of  $P$  and  $Q$  other than  $x$ . By considering those cases as we did in proving (iii), we obtain a contradiction.

(v) Suppose there exists a vertex  $y \in (V(P) \cap V(Q)) \setminus (\{x\} \cup N(D))$ . By (i) and (iv),  $y$  is an abstemious vertex since  $yPx'$  and  $yQx''$  are a pair of abstemious paths joining  $y$  and  $D$ . It is clear that  $\ell_2(y, D) < \ell_2(x, D)$ , contradicting the choice of  $x$ . This shows  $(V(P) \cap V(Q)) \setminus \{x\} \subseteq N(D)$ .

If  $|V(P) \cap V(Q) \cap N(D)| \geq 2$ , we take a vertex  $y$  from  $V(P) \cap V(Q) \cap N(D)$  such that  $\ell(xPy)$  is minimized. Since  $y \in N(D)$ ,  $y$  has a neighbor  $y'$  in  $D$ . Then  $P' = xPy'y'$  and  $Q$  are a pair of abstemious path joining  $x$  and  $D$  with  $\ell(P') + \ell(Q) < \ell(P) + \ell(Q)$ , a contradiction.

Now we show (vi). If  $V(P) \cap V(Q) = \emptyset$ , the result holds trivially. Otherwise, by (v),  $|V(P) \cap V(Q)| = 1$ . By the choice of  $P$  and  $Q$ , the vertex in  $V(P) \cap V(Q)$  must be adjacent to  $x'$  or  $x''$ . It follows that  $|(V(P) \cup V(Q)) \cap N(D)| \leq 2$ .  $\square$

Define a set  $I_1$  as follows: if  $k$  is even, a vertex  $v$  belongs to  $I_1$  if and only if  $v = x$  or  $v \in V(P) \cup V(Q)$  and  $d_G(v, D)$  is an even number less than  $k$ ; if  $k$  is odd, a vertex  $v$  belongs to  $I_1$  if and only if  $v \in V(P) \cup V(Q)$  and  $d_G(v, D)$  is an even number less than  $k$ . By the definition of  $I_1$  and (iv) of Claim 2,  $|I_1| = k-1$ .

**Claim 3**  $I_0 \cup I_1$  is an independent set of  $G$ .

*Proof* Since  $d_G(u, v) \geq 2$  for any  $u \in I_0$  and  $v \in I_1$ , it suffices to show that  $I_1$  is an independent set of  $G$ . Suppose that there exist vertices  $u$  and  $v$  of  $I_1$  adjacent in  $G$ . By the definition of  $I_1$ ,  $d_G(u, D) = d_G(v, D)$ . By (iv) of Claim 2,  $uPx'$  and  $uvQx''$  are a pair of abstemious edge-disjoint paths joining  $u$  and  $D$ . Thus  $u \in X$ , but  $\ell_2(u, D) < \ell_2(x, D)$ , which contradicts the choice of  $x$ . This shows that  $I_1$  is an independent set of  $G$ .  $\square$

**Claim 4**  $|((V(P) \cup V(Q)) \setminus D)| \leq 2|I_1| + 2$ .

*Proof* By (iii), (iv), (vi) of Claim 2,  $|((V(P) \cup V(Q)) \setminus D)| \leq 2k$ . So

$$|((V(P) \cup V(Q)) \setminus D)| \leq 2|I_1| + 2.$$

$\square$



Set  $D := V(C) \cup V(P) \cup V(Q)$  and  $I_0 := I_0 \cup I_1$ . It is clear that  $I_0 \subseteq D$ , and  $G[D]$  is 2-edge connected. If  $D$  is the dominating set, then  $D$  is 2-edge connected dominating set as we want. Otherwise, we repeat the above operation until  $D$  is a 2-edge connected dominating set. Since we add at least one vertex to  $I_0$  in each iteration, the procedure have to terminate when  $I_0$  become a maximal independent set of  $G$ . Let  $r$  be the number of the operations we implement. One can see that  $r \leq \alpha(G) - \alpha(C)$ . After the procedure terminate,  $D$  become a 2-edge connected dominating set of  $G$ . By Claim 4,

$$|D| \leq |C| + 2(\alpha(G) - \alpha(C)) + 2r \leq |C| + 4(\alpha(G) - \alpha(C)).$$

So we have

$$\gamma'_2(G) \leq |C| + 4(\alpha(G) - \alpha(C)) = \begin{cases} 4\alpha(G) - c, & \text{if } c \text{ is even;} \\ 4\alpha(G) - c + 2, & \text{if } c \text{ is odd.} \end{cases}$$

Since  $c \geq 3$ ,  $\gamma'_2(G) \leq 4\alpha(G) - 1$ . □

### 3 2-Connected dominating set

Let  $G$  be a graph, and let  $D \subseteq V(G)$  and  $x \in V(G) \setminus D$ . We call  $x$  *strongly abstemious* if there exist two edge-disjoint abstemious paths  $P$  and  $Q$  joining  $x$  and  $D$  such that  $V(P) \cap V(Q) = \{x\}$ . For a strongly abstemious vertex  $x$ , we use  $\ell'_2(x, D)$  to denote the minimum integer such that there exists a pair of edge-disjoint abstemious paths  $P$  and  $Q$  satisfying  $V(P) \cap V(Q) = \{x\}$  and  $\ell(P) + \ell(Q) = \ell'_2(x, D)$ .

*Proof of Theorem 3* We define  $C$ ,  $c$ ,  $D$ , and  $I_0$  as in the proof of Theorem 1. If  $D$  is a dominating set of  $G$ , then  $D$  is the dominating set as we desired. Otherwise, there exists at least one vertex  $x$  such that  $d(x, D) \geq 2$ . Let  $X$  be a set of strongly abstemious vertices which have distance at least 2 from  $D$ . By the similar argument as in the proof of Claim 1, one has

**Claim 5**  $X \neq \emptyset$ .

**Claim 6** Choose  $x \in X$  such that  $\ell'_2(x, D) = \min\{\ell'_2(v, D) : v \in X\}$ . Let  $P$  and  $Q$  be a pair of edge-disjoint abstemious paths joining  $x$  and  $D$  such that  $V(P) \cap V(Q) = \{x\}$  and  $\ell(P) + \ell(Q) = \ell'_2(x, D)$ . Denote by  $u$  and  $v$  the ends of  $P$  and  $Q$  other than  $x$ , and by  $u'$  and  $v'$  the neighbors of  $u$  and  $v$  on  $P$  and  $Q$ , respectively. If  $d_G(x, D) = k$ , then

- (i) For any  $y, z \in V(P)$ , if  $\ell(uPy) \leq \ell(uPz)$ , then  $d_G(y, D) \leq d_G(z, D)$ .
- (ii)  $(V(P) \setminus \{x\}) \cap N_G^k(D) = \emptyset$  or  $(V(Q) \setminus \{x\}) \cap N_G^k(D) = \emptyset$ .
- (iii) If  $(V(P) \setminus \{x\}) \cap N_G^k(D) \neq \emptyset$ , then  $|(V(P) \setminus \{x\}) \cap N_G^k(D)| = 1$  and the only vertex of  $(V(P) \setminus \{x\}) \cap N_G^k(D)$  is adjacent to  $x$  on  $P$ .
- (iv)  $|V(P) \cap N_G^i(D)| = 1$  and  $|V(Q) \cap N_G^i(D)| = 1$ , where  $2 \leq i < k$ .

(v)  $|V(P) \cap N(D)| = 1$  or  $|V(Q) \cap N(D)| = 1$ . If  $|V(P) \cap N(D)| \geq 2$ , each vertex of  $(V(P) \cap N(D)) \setminus \{u'\}$  has only neighbor  $v$  in  $D$ .

*Proof* (i), (ii), (iii) and (iv) in the claim can be proved by the similar argument as the proof of (i), (ii), (iii) and (iv) in Claim 2.

Next we prove (v).

Assume that  $|V(P) \cap N(D)| \geq 2$  and  $|V(Q) \cap N(D)| \geq 2$ . Take arbitrarily two vertices  $u'' \in (V(P) \cap N(D)) \setminus \{u'\}$  and  $v'' \in (V(Q) \cap N(D)) \setminus \{v'\}$ . By the choice of  $x$ ,  $P$  and  $Q$ , it is not hard to see that  $N(u'') \cap D = \{v\}$  and  $N(v'') \cap D = \{u\}$ . But,  $P' = xPu''v$  and  $Q' = xQv''u$  are a pair of abstemious paths joining  $x$  and  $D$ , a contradiction. This shows  $|V(P) \cap N(D)| = 1$  or  $|V(Q) \cap N(D)| = 1$ .

Assume that  $|V(P) \cap N(D)| \geq 2$ . Take arbitrarily a vertex  $u'' \in (V(P) \cap N(D)) \setminus \{u'\}$ . It suffices to show that  $N(u'') \cap D = \{v\}$ . By contradiction, suppose that  $u''$  has a neighbor  $w$  in  $D$  different from  $v'$ . Then  $P' = xPu''w$  and  $Q$  are a pair of abstemious paths joining  $x$  and  $D$ , a contradiction.  $\square$

We consider the following two cases.

*Case 1.*  $\ell'_2(x, D) > 6$ .

Let  $I_1$  be a maximum independent set of  $G[(V(P_1) \cup V(Q_1)) \setminus \{u, u', v, v'\}]$ . It is easy to check that  $|I_1| = \lceil \frac{\ell'_2(x, D) - 3}{2} \rceil = \lceil \frac{\ell'_2(x, D) - 1}{2} \rceil - 1$  by Claim 6. Set  $I_0 := (I_0 \setminus \{v\}) \cup I_1$  and  $D := D \cup V(P) \cup V(Q)$ . Therefore  $|I_0|$  increases by  $\lceil \frac{\ell'_2(x, D) - 1}{2} \rceil - 2$  and  $|D|$  increases by  $\ell'_2(x, D) - 1$ . It means that  $|D|$  increase by at most 6 when  $I_0$  increase by 1.

*Case 2.*  $\ell'_2(x, D) \leq 6$ .

Let  $I_1 = \{x\}$ . Set  $I_0 := I_0 \cup I_1$  and  $D := D \cup V(P) \cup V(Q)$ . It follows that  $|D|$  increase by at most 5 vertices when  $I_0$  increase by 1.

By the similar argument as in the proof of Claim 3,  $I_0$  is an independent set in  $G$ , and thus  $G[D]$  is 2-connected with  $I_0 \subseteq D$ . We will repeat the above operation until  $I_0$  become a maximal independent set in  $G$ , and thereby obtain a 2-connected dominating set  $D$  of  $G$ , with  $|D| \leq |C| + 6(\alpha(G) - \alpha(C))$ . So,

$$\gamma_2(G) \leq |C| + 6(\alpha(G) - \alpha(C)) = \begin{cases} 6\alpha(G) - 2c, & \text{if } c \text{ is even;} \\ 6\alpha(G) - 2c + 3, & \text{if } c \text{ is odd.} \end{cases}$$

Since  $c \geq 3$ ,  $\gamma_2(G) \leq 6\alpha(G) - 3$ .  $\square$

#### Proof of Theorem 4:

*Proof* We define  $C, c, D$ , and  $I_0$  as in the proof of Theorem 3. The symbols  $X, x, P, Q, u, v, u', v'$  have the same meanings as the statement of Claim 6, and let  $d_G(x, D) = k$ . Then the assertions of Claims 5 and 6 hold. Indeed, (v) of Claim 6 can be replaced with (v') below:

(v')  $|V(P) \cap N(D)| = 1$  or  $|V(Q) \cap N(D)| = 1$ , and if  $|V(P) \cap N(D)| \geq 2$ , then  $|V(P) \cap N(D)| = 2$ .

Suppose by contradiction that  $|V(P) \cap N(D)| > 2$ , let  $u_1, u_2 \in V(P) \cap N(D) \setminus \{u'\}$  such that  $u_1u_2 \in E(G)$ . By (v) of Claim 6, both  $u_1$  and  $u_2$  are adjacent to  $v$ . Thus  $u_1u_2vu_1$  is a triangle of  $G$ , which contradicts our assumption. Thus (v') holds.

Let  $u''$  be the neighbor of  $u'$  other than  $u$  on  $P$ . We consider the following two case.

*Case 1.*  $\ell'_2(x, D) > 6$ .

Let  $I_1$  be a maximum independent set of  $G[(V(P_1) \cup V(Q_1)) \setminus \{u, u', u'', v, v'\}]$ . It is easy to check that  $|I_1| = \lceil \frac{\ell'_2(x, D) - 4}{2} \rceil = \lceil \frac{\ell'_2(x, D)}{2} \rceil - 2$  by (v'). Set  $I_0 := I_0 \cup I_1$  and  $D := D \cup V(P) \cup V(Q)$ . Therefore,  $|I_0|$  increase by  $\lceil \frac{\ell'_2(x, D)}{2} \rceil - 2$  and  $|D|$  increase by  $\ell'_2(x, D) - 1$ . That is,  $|D|$  increase by at most 5 while if  $|I_0|$  increase by 1.

*Case 2.*  $\ell'_2(x, D) \leq 6$ .

Let  $I_1 = \{x\}$ . Set  $I_0 := I_0 \cup I_1$  and  $D := D \cup V(P) \cup V(Q)$ . Thus  $|D|$  increase by at most 5 while if  $|I_0|$  increase by 1.

In each of the above two cases, we can check that  $I_0$  is an independent set in  $G$  and  $G[D]$  is 2-connected. As we did in the proof of Theorem 1, We repeat the above operation until  $D$  is a dominating set of  $G$ . The number  $r$  of iteration is at most  $\alpha(G) - \alpha(C)$ , and thus  $|D| \leq |C| + 5r$  vertices. So we have

$$\gamma_2(G) \leq |C| + 5(\alpha(G) - \alpha(C)) = \begin{cases} 5\alpha(G) - \frac{3c}{2}, & \text{if } c \text{ is even;} \\ 5\alpha(G) - \frac{3c-1}{2} + 2, & \text{if } c \text{ is odd.} \end{cases}$$

Since  $G$  is triangle-free,  $c \geq 4$ . By taking  $c = 4$  and  $c = 5$  into the left side on the above inequality, one has  $\gamma_2(G) \leq 5\alpha(G) - 5$ .  $\square$

#### 4 Concluding remarks

In Proposition 1, we show a necessary and sufficient condition for a graph having 2-edge connected dominating set.

A basic problem is to find a simple necessary and sufficient condition for a graph having a  $k$ -edge connected (resp.  $k$ -connected) dominating set for  $k \geq 3$ .

Based on our main theorems, we propose the following two problems:

- (1) is there a set  $S' \subseteq V(G) \setminus S$  such that  $|S'| \leq 4|S|$  and  $G[S \cup S']$  is 2-edge connected for any maximal independent set  $S$  of a 2-edge connected graph  $G$ ?
- (2) is there a set  $S' \subseteq V(G) \setminus S$  such that  $|S'| \leq 6|S|$  and  $G[S \cup S']$  is 2-connected for any maximal independent set  $S$  of a 2-connected graph  $G$ ?

**Acknowledgments** This work is supported by NSFC (11161046), Xinjiang Young Talent Project (2013721012), and Research Found of Henan Normal University (qd13042). The authors are grateful to the referees for their careful reading and valuable comments.

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